Lesson 3: Ohm's Law

ChatGPT edition — a graph is stronger than a slogan

Measure path, current, voltage, and power

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

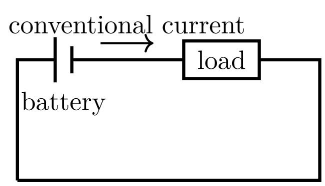
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- 2–4 AA-cell switched holder
- 100, 220, and 470 Ω resistors
- LED with series resistor
- Digital multimeter

Predict first

For the same source, which resistor should permit the greatest current?



Investigate

1. With power off, build source–resistor–switch in one closed path. Adult checks meter jack and range.
2. Place the ammeter in the path; record current. Disconnect.
3. Put the voltmeter across the resistor; reconnect briefly and record voltage.
4. Repeat with three resistors. Calculate $R = V/I$ and $P = VI$; graph V against I.

Explain from the evidence

Current is charge per time; voltage describes energy transfer per charge. For an ohmic part at stable temperature, $V = IR$. A voltmeter compares two points and an ammeter joins the current path. Putting an ammeter directly across a source makes a short circuit. Power $P = VI$ tracks energy transfer rate, not a substance consumed by the first component.

Change one thing

Add a second resistor in series, then in parallel. Predict equivalent resistance before measuring.

Evidence record	
Prediction	______
One change	______
Observation	______
Claim + because	______

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.